



LEARNING ROADMAP

Data Science Roadmap

From Python Basics to Production-Ready ML Models

Intensive

40-50 hrs/week

Balanced

15-20 hrs/week

Flexible

8-10 hrs/week

■ 7 Phases ★ Beginner to Advanced

<https://crio.do>

Overview

This roadmap will guide you through the complete data science journey. You'll build a strong foundation in Python and statistics, understand the mathematics that powers machine learning (and why it matters), master data analysis and visualization, and learn to build, evaluate, and deploy ML models. By the end, you'll be equipped to solve real-world problems with data.

Who is this for?

Aspiring data scientists, analysts looking to upskill, software engineers transitioning to ML, students in STEM fields, or anyone curious about making sense of data and building intelligent systems.

Prerequisites

- Basic computer literacy
- High school mathematics (comfortable with algebra)
- Curiosity about patterns in data
- Willingness to think analytically
- A computer with Python installed

What you'll learn

- Write Python code for data manipulation and analysis
- Apply statistical methods to draw insights from data
- Understand the math behind ML algorithms (not just use them as black boxes)
- Build and evaluate machine learning models
- Create compelling data visualizations
- Deploy models to production
- Communicate findings to stakeholders

Learning Plans

Choose a plan that fits your schedule. All plans cover the same content.

Intensive

Fast Track

40-50

hrs/week

**240-
300**

total hrs

Full-time immersive learning. Ideal for career changers who can dedicate themselves completely to this transition.

Ideal for:

- Career changers with dedicated time
- Recent graduates
- Those preparing for DS interviews
- Bootcamp-style learners

Balanced

Recommended

15-20

hrs/week

**270-
360**

total hrs

Balanced pace allowing deep understanding while maintaining other commitments. The recommended path for most learners.

Ideal for:

- Working professionals
- Students with coursework
- Those who prefer thorough understanding
- Self-paced learners

Flexible

Self-paced

8-10

hrs/week

**288-
360**

total hrs

Flexible learning for those with limited availability. More time to absorb complex mathematical concepts.

Ideal for:

- Full-time employees
- Parents and caregivers
- Those learning multiple skills
- Weekend learners

Phase Schedule by Plan

Phase	Topic	Intensive	Balanced	Flexible
1	Python Programming Foundation	Week 1	Weeks 1-3	Weeks 1-5
2	Statistics & Probability	Week 2	Weeks 4-6	Weeks 6-11
3	Mathematics for Machine Learning	Weeks 3-4	Weeks 7-10	Weeks 12-19
4	Data Analysis & Visualization	Week 5	Weeks 11-13	Weeks 20-25
5	Machine Learning	Weeks 6-7	Weeks 14-18	Weeks 26-35
6	Deep Learning & Specialization	Week 8	Weeks 19-22	Weeks 36-43
7	Production & Career	Week 9	Weeks 23-26	Weeks 44-52

1 Python Programming Foundation

Intensive: Week 1

Balanced: Weeks 1-3

Flexible: Weeks 1-5

Python is the lingua franca of data science. Before diving into statistics and ML, you need to be comfortable writing Python code, working with data structures, and using the core libraries that power the data science ecosystem.

Learning Objectives

- Write clean, readable Python code
- Work with lists, dictionaries, and other data structures
- Use functions and classes effectively
- Navigate Jupyter notebooks fluently
- Install and manage packages with pip/conda
- Understand Python's role in the data ecosystem

Topics to Cover

Python Fundamentals

Core language features

- Variables & Data Types
- Control Flow (if/else, loops)
- Functions & Lambda
- List Comprehensions
- Error Handling

Data Structures

Organizing and storing data

- Lists & Tuples
- Dictionaries & Sets
- Strings & String Methods
- Working with Files
- JSON Handling

Development Environment

Tools of the trade

- Jupyter Notebooks
- VS Code Setup
- Virtual Environments
- pip & conda
- Git Basics

Python for Data

Introduction to data libraries

- NumPy Basics
- Pandas Introduction
- Reading CSV/Excel Files
- Basic Data Inspection
- Simple Plots with Matplotlib

Hands-on Projects

Data File Processor

Build a CLI tool that reads CSV files, performs basic cleaning (removing duplicates, handling missing values), and outputs summary statistics

File I/O

Data Structures

Functions

Error Handling

Personal Expense Analyzer

Create a notebook that analyzes your bank statement or expense data, categorizes spending, and visualizes trends over time

Pandas

Data Cleaning

Matplotlib

Jupyter

Resources

DOCUMENTATION

[Python Official Tutorial](#)

BOOK

[Automate the Boring Stuff with Python](#)

TUTORIAL

[Real Python Tutorials](#)

Milestone



Complete both projects. You should be able to write Python scripts that read, process, and analyze data files without constantly Googling basic syntax.

2 Statistics & Probability

Intensive: Week 2

Balanced: Weeks 4-6

Flexible: Weeks 6-11

Statistics is the backbone of data science. It's how we make sense of data, quantify uncertainty, and make decisions with confidence. This isn't just theory – every concept here directly applies to real data science work. Skip this phase at your peril; without statistics, you're just guessing.

Learning Objectives

- Summarize datasets using descriptive statistics
- Understand probability and its role in prediction
- Work with different probability distributions
- Conduct and interpret hypothesis tests
- Quantify relationships with correlation and covariance
- Apply Bayes' theorem to update beliefs with data

Topics to Cover

Descriptive Statistics

WHY: Before building any model, you must understand your data. Descriptive stats help you spot outliers, understand distributions, and identify data quality issues.

- Mean, Median, Mode
- Variance & Standard Deviation
- Percentiles & Quartiles
- Skewness & Kurtosis
- Data Visualization for Stats

Probability Fundamentals

WHY: Data science is about making predictions under uncertainty. Probability gives us the language and tools to quantify 'how likely' something is.

- Basic Probability Rules
- Conditional Probability
- Independence
- Bayes' Theorem
- Expected Value

Probability Distributions

WHY: Real-world phenomena follow patterns (distributions). Heights are normal, customer arrivals are Poisson, click-through rates are binomial. Knowing distributions lets you model reality.

- Normal Distribution
- Binomial & Bernoulli
- Poisson Distribution
- Uniform Distribution
- Central Limit Theorem

Inferential Statistics

WHY: You rarely have all the data. Inference lets you draw conclusions about populations from samples – essential for A/B testing, surveys, and any real-world analysis.

- Sampling & Sampling Distributions
- Confidence Intervals
- Hypothesis Testing
- p-values & Significance
- t-tests & Chi-square Tests

Hands-on Projects

A/B Test Analyzer

Given data from a website A/B test (conversion rates for control vs treatment), determine if the difference is statistically significant and calculate the confidence interval

Hypothesis Testing

Confidence Intervals

Statistical Significance

Python scipy.stats

Sales Data Statistical Report

Analyze a retail dataset: compute descriptive statistics by category, identify correlations between variables, test hypotheses about seasonal effects

Descriptive Statistics

Correlation

Distribution Fitting

Statistical Visualization

Resources

BOOK

Think Stats (Free Book)

COURSE

Khan Academy Statistics

INTERACTIVE

Seeing Theory (Visual Stats)

Milestone



Complete the A/B Test Analyzer project. You should be able to look at any dataset and immediately compute relevant statistics, identify distributions, and make statistically-grounded conclusions.

3 Mathematics for Machine Learning

Intensive: Weeks 3-4

Balanced: Weeks 7-10

Flexible: Weeks 12-19

This is where many learners struggle or skip entirely – and it's why they later hit a ceiling. Understanding the math doesn't mean deriving proofs; it means knowing WHY algorithms work, WHEN they'll fail, and HOW to debug them. This knowledge separates practitioners who can only call sklearn functions from those who can actually solve novel problems.

Learning Objectives

- Understand vectors, matrices, and their operations
- Grasp how linear transformations relate to ML
- Know why gradient descent works (calculus)
- Connect probability theory to model uncertainty
- Recognize the math when reading ML papers or documentation

Topics to Cover

Linear Algebra

WHY: Machine learning IS linear algebra. Your data is a matrix. Features are vectors. Neural networks are matrix multiplications. PCA uses eigenvalues. Recommendation systems use matrix factorization. Without this, ML is a black box.

- Vectors & Vector Operations
- Matrices & Matrix Multiplication
- Linear Transformations
- Eigenvalues & Eigenvectors
- Matrix Decomposition (SVD, PCA)

Calculus for Optimization

WHY: How does a model 'learn'? It minimizes a loss function using gradient descent – which is calculus. Understanding derivatives tells you why learning rates matter, why models get stuck, and how to fix them.

- Derivatives & Gradients
- Partial Derivatives
- Chain Rule (Backpropagation!)
- Gradient Descent Intuition
- Convexity & Local Minima

Probability for ML

WHY: ML models don't just predict – they estimate probabilities. Understanding MLE helps you grasp logistic regression. Bayesian thinking enables uncertainty quantification. This is how you know when your model is confident vs guessing.

- Maximum Likelihood Estimation
- Bayesian Inference Basics
- Prior, Likelihood, Posterior
- Probabilistic Graphical Models (intro)
- Information Theory Basics

Hands-on Projects

Implement Linear Regression from Scratch

Build linear regression using only NumPy – implement the normal equation AND gradient descent. Visualize how gradient descent converges. Compare with sklearn's implementation.

Matrix Operations

Gradient Descent

Loss Functions

NumPy

PCA Visualization Tool

Implement PCA from scratch using eigendecomposition. Apply it to a high-dimensional dataset (like faces or MNIST) and visualize the principal components.

Eigenvalues/Eigenvectors

Dimensionality Reduction

Matrix Decomposition

Visualization

Resources

VIDEO 3Blue1Brown: Essence of Linear Algebra

VIDEO 3Blue1Brown: Essence of Calculus

BOOK Mathematics for Machine Learning (Book)

COURSE Khan Academy Linear Algebra

Milestone



Implement linear regression from scratch using gradient descent. You should be able to explain WHY the gradient points toward the minimum, WHAT eigenvalues represent, and HOW probability connects to model training.

4 Data Analysis & Visualization

Intensive: Week 5

Balanced: Weeks 11-13

Flexible: Weeks 20-25

Now we combine your Python skills, statistical knowledge, and mathematical foundation into practical data analysis. This is the bread and butter of data science – exploring datasets, cleaning messy data, engineering features, and creating visualizations that tell stories.

Learning Objectives

- Master Pandas for data manipulation
- Handle missing data, outliers, and data quality issues
- Perform exploratory data analysis (EDA) systematically
- Create publication-quality visualizations
- Engineer features that improve model performance
- Work with different data formats (CSV, JSON, SQL, APIs)

Topics to Cover

Pandas Mastery

The workhorse of data analysis

- DataFrames & Series
- Indexing & Selection
- Groupby & Aggregation
- Merging & Joining
- Pivot Tables
- Time Series in Pandas

Data Cleaning

80% of data science is cleaning data

- Handling Missing Values
- Outlier Detection & Treatment
- Data Type Conversions
- String Cleaning & Regex
- Deduplication
- Data Validation

Exploratory Data Analysis

Understanding data before modeling

- Univariate Analysis
- Bivariate Analysis
- Correlation Analysis
- Distribution Analysis
- Automated EDA Tools
- Documenting Findings

Data Visualization

Communicating insights visually

- Matplotlib Deep Dive
- Seaborn for Statistical Plots
- Plotly for Interactive Viz
- Choosing the Right Chart
- Dashboard Basics
- Visualization Best Practices

Hands-on Projects

End-to-End EDA: Titanic or Housing Dataset

Perform comprehensive EDA on a classic dataset. Document every finding, handle all data quality issues, create visualizations for each insight, and prepare the data for modeling.

Pandas

Data Cleaning

Statistical Analysis

Visualization

Documentation

Interactive Dashboard

Build an interactive dashboard using Plotly Dash or Streamlit that allows users to explore a dataset, filter by dimensions, and view dynamic visualizations

Plotly/Streamlit

Interactive Viz

User Interface

Data Filtering

Resources

DOCUMENTATION

[Pandas Documentation](#)

BOOK

[Python Data Science Handbook](#)

TUTORIAL

[Seaborn Tutorial](#)

Milestone



Complete a full EDA project with a written report. You should be able to take any messy dataset and, within hours, have it cleaned, analyzed, and visualized with key insights documented.

5 Machine Learning

Intensive: Weeks 6-7

Balanced: Weeks 14-18

Flexible: Weeks 26-35

This is what you've been building toward. With your foundation in Python, statistics, and math, you'll now understand ML algorithms deeply – not just how to call them, but why they work, when to use them, and how to evaluate them properly.

Learning Objectives

- Understand the ML workflow end-to-end
- Master supervised learning (regression & classification)
- Apply unsupervised learning (clustering & dimensionality reduction)
- Evaluate models properly (avoiding common pitfalls)
- Tune hyperparameters effectively
- Handle imbalanced data and other real-world challenges

Topics to Cover

ML Fundamentals

Core concepts that apply everywhere

- Supervised vs Unsupervised
- Train/Validation/Test Split
- Bias-Variance Tradeoff
- Overfitting & Underfitting
- Cross-Validation
- Feature Scaling

Regression Algorithms

Predicting continuous values

- Linear Regression (revisited)
- Polynomial Regression
- Regularization (Ridge, Lasso)
- Decision Tree Regression
- Random Forest Regression
- Gradient Boosting (XGBoost)

Classification Algorithms

Predicting categories

- Logistic Regression
- Decision Trees
- Random Forests
- Support Vector Machines
- K-Nearest Neighbors
- Naive Bayes
- Gradient Boosting Classifiers

Unsupervised Learning

Finding patterns without labels

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN
- PCA (applied)
- t-SNE & UMAP
- Anomaly Detection

Model Evaluation & Selection

Knowing if your model is actually good

- Regression Metrics (MSE, MAE, R^2)
- Classification Metrics (Accuracy, Precision, Recall, F1)
- ROC-AUC & PR Curves
- Confusion Matrix Analysis
- Hyperparameter Tuning
- Model Selection Strategies

Hands-on Projects

House Price Prediction

Build a complete ML pipeline for predicting house prices. Compare multiple algorithms, perform feature engineering, tune hyperparameters, and create a final report with model interpretation.

Regression

Feature Engineering

Model Comparison

Hyperparameter Tuning

Customer Churn Prediction

Predict which customers will churn using classification. Handle class imbalance, interpret feature importance, and provide business recommendations based on your model.

Classification

Imbalanced Data

Model Interpretation

Business Insights

Customer Segmentation

Use clustering to segment customers based on behavior. Determine optimal number of clusters, profile each segment, and visualize the results.

Clustering

Unsupervised Learning

Segment Profiling

Visualization

Resources

DOCUMENTATION

[Scikit-learn Documentation](#)

BOOK

[Hands-On Machine Learning \(Aurélien Géron\)](#)

VIDEO

[StatQuest ML Playlist](#)

Milestone



Complete all three projects with documented notebooks. You should be able to take a business problem, frame it as an ML task, build and evaluate models, and communicate results to non-technical stakeholders.

6

Deep Learning & Specialization

Intensive: Week 8

Balanced: Weeks 19-22

Flexible: Weeks 36-43

Deep learning has revolutionized AI. While not every problem needs neural networks, understanding them opens doors to computer vision, NLP, and cutting-edge applications. Choose a specialization based on your interests.

Learning Objectives

- Understand neural network fundamentals
- Build models with TensorFlow or PyTorch
- Apply transfer learning for quick results
- Choose and explore a specialization (NLP, CV, or Time Series)
- Know when deep learning is appropriate vs overkill

Topics to Cover

Neural Network Fundamentals

How deep learning actually works

- Perceptrons & Activation Functions
- Forward & Backpropagation
- Loss Functions & Optimizers
- Regularization (Dropout, BatchNorm)
- TensorFlow/Keras Basics
- PyTorch Basics

Computer Vision Track

If you're interested in image/video

- Convolutional Neural Networks
- Image Classification
- Object Detection
- Transfer Learning (ResNet, VGG)
- Data Augmentation
- OpenCV Basics

NLP Track

If you're interested in text/language

- Text Preprocessing
- Word Embeddings
- RNNs & LSTMs
- Transformers & Attention
- BERT & GPT (using, not training)
- Hugging Face Ecosystem

Time Series Track

If you're interested in forecasting

- Time Series Decomposition
- ARIMA Models
- Prophet
- LSTMs for Time Series
- Feature Engineering for Time
- Evaluation Metrics for Forecasting

Hands-on Projects

Image Classifier with Transfer Learning

Build a custom image classifier using a pretrained model (ResNet/VGG). Fine-tune on your own dataset and deploy as a simple web app.

CNNs

Transfer Learning

Data Augmentation

Model Deployment

Sentiment Analysis Pipeline

Build an NLP pipeline that classifies sentiment in product reviews. Compare traditional ML with transformer-based approaches.

Text Processing

Transformers

Model Comparison

NLP

Resources

COURSE

Fast.ai Practical Deep Learning

COURSE

Deep Learning Specialization (Coursera)

COURSE

Hugging Face Course

Milestone



Complete one specialization track project. You should understand when deep learning adds value and be able to implement solutions using modern frameworks and pretrained models.

7 Production & Career

Intensive: Week 9

Balanced: Weeks 23-26

Flexible: Weeks 44-52

The final step: taking your models from notebooks to production and yourself from learner to professional. This phase covers deployment, MLOps basics, and building the portfolio that gets you hired.

Learning Objectives

- Deploy ML models as APIs
- Understand MLOps fundamentals
- Build a compelling portfolio
- Prepare for data science interviews
- Contribute to the DS community

Topics to Cover

Model Deployment

Getting models into production

- Flask/FastAPI for ML APIs
- Docker Basics
- Cloud Deployment (AWS/GCP/Azure)
- Streamlit for ML Apps
- Model Serialization
- API Best Practices

MLOps Fundamentals

Maintaining ML systems

- ML Pipelines
- Experiment Tracking (MLflow)
- Model Versioning
- Monitoring & Drift Detection
- CI/CD for ML
- Data Versioning (DVC)

Portfolio & Career

Getting hired

- GitHub Portfolio Best Practices
- Project Documentation
- Technical Blogging
- LinkedIn Optimization
- Interview Preparation
- Networking in DS Community

Hands-on Projects

End-to-End ML Application

Take one of your previous projects and deploy it as a full web application. Include data pipeline, model training, API, and frontend. Document everything.

Full Stack ML

Deployment

Documentation

DevOps Basics

Portfolio Website

Create a portfolio showcasing your best 3-5 projects. Each project should have a clear problem statement, methodology, results, and lessons learned.

Communication

Documentation

Personal Branding

Web Development

Resources

COURSE

Made With ML (MLOps)

COURSE

Full Stack Deep Learning

BOOK

ML System Design Interview Book

Milestone



Deploy at least one ML application publicly and have a portfolio with 3-5 documented projects. You should be able to discuss your projects in depth and demonstrate end-to-end ML capabilities.

Your Path Forward

Career Opportunities

✓ Data Scientist

✓ Machine Learning Engineer

✓ Data Analyst

✓ Business Intelligence Analyst

✓ Research Scientist

✓ MLOps Engineer

Tips for Success

- Don't skip the math – it's what separates good data scientists from great ones
- Build projects with real, messy data – Kaggle competitions are good but not enough
- Learn to communicate findings to non-technical audiences
- Statistics before machine learning, always
- Understand algorithms deeply before using AutoML
- Join communities: Kaggle, Reddit r/datascience, Twitter DS community
- Read papers, but focus on understanding intuition over mathematical rigor
- Contribute to open source or write about what you learn
- Practice explaining your projects – interviews are about communication
- Stay curious and keep learning – the field evolves rapidly

Next Steps

1. Set up your Python environment (Anaconda recommended)
2. Choose your learning pace (intensive, standard, or part-time)
3. Start with Phase 1 – don't skip ahead
4. Join a community for accountability and help
5. Begin building your GitHub profile from day one
6. Commit to consistency over intensity

Ready to Start Your Journey?

Everything in this roadmap is free to learn on your own. But if you'd like structured guidance, hands-on projects, and mentor support – we'd love to have you at Crio.Do

Explore Crio.Do

<https://crio.do>